

Module 4

Chapter 3

Conditional Probability

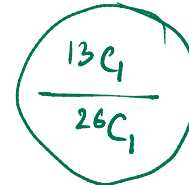
Introduction to Conditional probability

Sometimes the probability of an event changes as we gain knowledge about a related event that has occurred (or is guaranteed to occur) or by some additional conditions imposed in the experiment.

This new probability is called **Conditional Probability**, because we have some prior information about conditions.

Let's take an example

A card is drawn at random from a deck of 52 cards:


$$\frac{13C_1}{26C_1}$$

- Let ***H*** be the event that a ***heart*** is drawn.
- Let ***R*** be the event that a ***red*** card is drawn.
- Let ***F*** be the event that a ***Face*** card is drawn (King, queen and Jack)

a) If I draw a card at random, $P(H) = \frac{13}{52}$

b) Now, without showing you the card in my hand, if I tell you that the card in my hand is a ***Red*** card, What is the probability that it is a heart card? Because you know that the card drawn is red, the ***Sample Space*** of all possible outcomes has size **26** (there are **26 red cards**), **not 52**. Of the 26 red cards, 13 are hearts, so there are 13 successful outcomes. Our probability now becomes: $\frac{13}{26}$

c) If I draw a card at random, $P(F) = \frac{12}{52}$

$$\frac{6C}{26C} = \frac{6}{26}$$

d) If I draw a card at random, and without showing you the card, I tell you that the card is **red**, then what are the chances that it is a **face card** (i.e. what is $P(F|R)$)?

There are **6 red face cards** and **26 red cards**, so $P(F|R) = 6/26 = 3/13$.

**Notice that in this case, the probability does not change even though both the sample space and the event space do change.

We can now go ahead and define conditional probability.

Probability of an **event A**, **given** that **event B** has occurred, is given by:

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}} = \frac{P(A \cap B)}{P(B)}$$

This is called the **conditional probability of A given B**.

Let's look at an example

Here is the data I recorded over the last 30 days:

S: Number of days the weather was sunny.

NS: Number of days the weather was not sunny.

G: Number of days my mood was good.

NG: Number of days my mood was not good.

	S	NS
G	9	6
NG	1	14

- ***If I pick a day at random from the above data, what is the probability that I was in a good mood that day?***

We know, total number of days the data was recorded over was 30, and from the above table, if we look at the row marked G, we have $n(G) = (9 + 6) = 15$

days. So, $P(G) = \frac{15}{30} = \frac{1}{2}$

- *What is the probability that a chosen day was sunny?*

Let's look at the column marked S: we have $n(S) = (9 + 1) = 10$ sunny days. So,

$$P(S) = \frac{10}{30} = \frac{1}{3}$$

- *What is $P(G|S)$?*

We know, $P(G|S) = \frac{n(G \cap S)}{n(S)} = \frac{9}{10}$ (We can get $n(G \cap S)$ by looking at the cell corresponding to S and G.)

- *What is $P(S|G)$?*

We use the same method as above and get: $P(S|G) = \frac{n(G \cap S)}{n(G)} = \frac{9}{15}$

Conditional probability and independence

Sometimes $P(A|B) = P(A)$. This simply means that the fact that B has occurred, doesn't influence probability of A. In other words, this means that **A and B are independent**. Similarly, when A and B are independent, $P(B|A) = P(B)$.

We can make use of **conditional probabilities** in thinking about the **conjunctions of events that aren't independent**.

Let's take an example

$$P(F_2 | F_1) = 0.5$$

Suppose we know that if a server fails on day 1, it has a 0.5 chance of failing on day 2. On a normal day it has a 0.01 chance of failure. What is the chance of two failures in a row starting from a given day?

We know,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Rearranging the terms, we get

$$P(A \cap B) = P(A|B) \times P(B)$$

Let the event a server fails on day 1 be F_1

$$P(F_1) = 0.01$$

$$P(F_2 \cap F_1) = ?$$

$$P(F_2 | F_1) = \frac{P(F_2 \cap F_1)}{P(F_1)}$$

$$\underline{P(F_2 \cap F_1)} = \frac{P(F_2 | F_1)}{0.5} \times \frac{P(F_1)}{0.01}$$

Let the event a server fails on day 2 be F_2

Given,

$$P(F_2|F_1) = 0.5$$

$$P(F_1) = 0.01$$

To find, $P(F_1 \cap F_2)$

$$P(F_1 \cap F_2) = P(F_2|F_1) \times P(F_1) = 0.5 \times 0.01 = 0.005$$

Introduction to Bayes' Rule

Suppose you have a case where you know the probability of symptoms given a particular disease, i.e. $Pr(\text{Symptom}|\text{Disease})$, but you are interested in finding the probability that a patient has a certain disease given the symptoms, i.e.

$Pr(\text{Disease}|\text{Symptom})$. We can use Bayes' Rule for this. Bayes' Rule gives us a formula for *flipping conditional probabilities*.

We can fairly easily *derive Bayes' Rule* from the formula for *conditional probability*:

We know:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B) \times P(B) \dots \dots \dots (1)$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A \cap B) = P(B|A) \times P(A) \dots \dots \dots (2)$$

From equation (1) and (2), we get:

$$P(A|B) \times P(B) = P(B|A) \times P(A)$$

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

And

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$
$$P(A|B) \cdot P(B) = P(A \cap B)$$

$$P(B|A) \cdot P(A) = P(A \cap B)$$

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$$

Bayes' Rule

$$P(B|A) = \frac{P(A|B) \times P(B)}{P(A)}$$

In general, we can define the formula for Bayes' Rule as:

$$P(\text{Hypothesis}|\text{Evidence}) = \frac{P(\text{Evidence}|\text{Hypothesis}) \times P(\text{Hypothesis})}{P(\text{Evidence})}$$

Let's look at an example

Suppose 1 in 10000 people is a criminal on the FBI's most wanted list. A new airport face recognition system has a 0.97 probability of correctly identifying someone on the wanted list. But it has a 0.01 probability of a false alarm on a normal person. If the detector goes off, what is the chance it's a criminal?

Let's first write every piece of relevant information we have, in terms of mathematical notations:

Given:

$$P(\text{Criminal}) = \frac{1}{10000}$$

$$P(\text{Alarm}|\text{Criminal}) = \frac{97}{100}$$

$$P(\text{Alarm}|\text{Not Criminal}) = \frac{1}{100}$$

$$\begin{aligned} P(\text{alarm}) &= P(\text{alarm} \cap \text{Criminal}) \\ &\quad + P(\text{alarm} \cap \text{Not Criminal}) \end{aligned}$$

To find:

$$P(\text{Criminal}|\text{Alarm}) = ?$$

Using Bayes' Rule, we know

$$P(\text{Criminal}|\text{Alarm}) = \frac{P(\text{Alarm}|\text{Criminal}) \times P(\text{Criminal})}{P(\text{Alarm})}$$

$A \rightarrow \text{alarm}$
 $B \rightarrow \text{Criminal}$

$$\frac{P(A \cap B)}{P(B)} = P(A|B)$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

We have everything we need except $P(\text{Alarm})$.

But we know:

$$P(\text{alarm}) = P(\text{Alarm} \cap \text{Criminal}) + P(\text{Alarm} \cap \text{Not Criminal})$$

$$P(\text{Alarm} \cap \text{Criminal}) = P(\text{Alarm}|\text{Criminal}) \times P(\text{Criminal}) = \frac{97}{100} \times \frac{1}{1000}$$

$$P(\text{Alarm} \cap \text{Not Criminal}) = P(\text{Alarm}|\text{Not Criminal}) \times P(\text{Not Criminal})$$

$$= \frac{1}{100} \times \left(1 - \frac{1}{1000}\right)$$

$$(1 - P(\text{Criminal}))$$

Putting everything back into the formula for Bayes' Rule:

$$P(\text{Criminal}|\text{Alarm}) = \frac{P(\text{Alarm}|\text{Criminal}) \times P(\text{Criminal})}{P(\text{Alarm})}$$

$$P(\text{Criminal}|\text{Alarm}) = \frac{\frac{97}{100} \times \frac{1}{1000}}{\left\{ \left(\frac{97}{100} \times \frac{1}{1000} \right) + \left(\frac{1}{100} \times \left(1 - \frac{1}{1000} \right) \right) \right\}} = 0.0096$$

Let's Practice

$$P(H) = 0.003 \quad Y \rightarrow \text{Test +ve} \quad P(Y|H) = 0.99$$

$$P(H|Y) \quad H \rightarrow \text{HIV} \quad P(N|NH) = 0.99$$

- The results of a survey of a group of 100 people having insurances with a certain company are as follows: 40% have both home and car insurances with the company. The probability that a person selected at random from this group, has a car insurance is 0.7. What is the probability that a person selected at random has a home insurance knowing that they have a car insurance?
- Suppose the test for HIV is 99% accurate in both directions and 0.3% of the population is HIV positive. If someone tests positive, what is the probability that they actually are HIV positive?
- In a group of kids, if one is selected at random, the probability that they like oranges is 0.6, the probability that they like oranges and apples is 0.3. If a kid who likes oranges is selected at random, what is the probability they also like apples?

- A group of 200 students were asked whether they play football or basketball. Among the group, 120 said they played football, 50 said they played basketball and 20 said they played both.
 - a) What is the probability that a student selected at random plays football, given that he plays basketball?
 - b) What is the probability that a student plays basketball, given that he plays football?
 - c) What is the probability that a student plays football, given that he plays one game only?

1. Let $H \rightarrow$ Home insurance
 $C \rightarrow$ Car insurance

Given, $P(C) = 0.7$
 $P(C \cap H) = 0.4$

To find, we know, $P(H|C) = \frac{P(H \cap C)}{P(C)} = \frac{0.4}{0.7} = 0.57$

$$2. P(T_P | H_P) = 0.99$$

$$P(T_N | H_N) = 0.99$$

$$P(H_P) = 0.003$$

$H_N \rightarrow$ Person is actually HIV negative

$T_N \rightarrow$ Person tests negative

$H_P \rightarrow$ Person is actually HIV positive

$T_P \rightarrow$ Person tests positive

To find,
$$P(H_P | T_P) = \frac{P(T_P | H_P) \cdot P(H_P)}{P(T_P)}$$

$$= \frac{0.99 \times 0.003}{(0.99 \times 0.003) + (1 - 0.99)(1 - 0.003)}$$

$$P(T_P) = P(T_P \cap H_P) + P(T_P \cap H_N)$$

$$= P(T_P | H_P)P(H_P) + P(T_P | H_N) \cdot P(H_N)$$

3. Let $O \rightarrow$ event kid likes Oranges
 $A \rightarrow$ event kid likes apples

given $P(O) = 0.6$

$$P(A \cap O) = 0.3$$

To find $P(A|O)$

$$P(A|O) = \frac{P(A \cap O)}{P(O)} = \frac{0.3}{0.6} = 0.5$$

$$3. \quad P(F) = 120/200 = 0.6$$

$$P(B) = 50/200 = 0.25$$

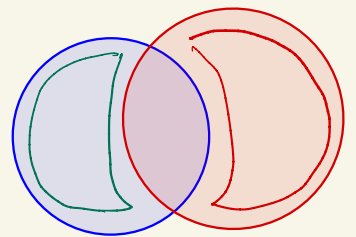
$$P(F \text{ and } B) = 20/200 = 0.1$$

$$a) \quad P(F|B) = \frac{P(F \cap B)}{P(B)} = \frac{0.1}{0.25} = 0.4$$

$$b) \quad P(B|F) = \frac{P(F \cap B)}{P(F)} = \frac{0.1}{0.6} = 0.17$$

c) Let $O \rightarrow$ event that a student plays only one game

$$n(O) = \underbrace{(120 - 20)}_{n(F \cap O)} + \underbrace{(50 - 20)}_{n(B \cap O)} = 130$$



$$n(F \cap O) = 120 - 20 = 100$$

$$\therefore P(F|O) = \frac{n(F \cap O)}{n(O)} = \frac{100}{130} = 0.77$$